

A Counterexample to Conjecture 00000002058: Matrix Algebras Are Equationally Noetherian

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000002058 asserts that algebraic sets over the matrix algebra $M_n(K)$ are Noetherian if and only if $n \leq 2$, and in particular that $M_3(K)$ is not equationally Noetherian. In fact $M_n(K)$ is equationally Noetherian for every n and every field K . Matrix equations are, entrywise, polynomial equations in the entries of the unknowns, so Hilbert's basis theorem applies. The argument is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000002058 reads:

Definition: Equational Noetherianity means every finite system of equations is equivalent to a finite subsystem. Conjecture: Algebraic sets over $M_n(K)$ are Noetherian if and only if $n \leq 2$; concretely, algebraic sets over $M_3(K)$ are non-Noetherian (by a construction of infinite ascending chains of commutator polynomials).

2 Reading of the statement

The definition as printed (“every *finite* system ... is equivalent to a finite subsystem”) is satisfied by every algebra, since a finite system is a finite subsystem of itself. Read that way, $M_3(K)$ is trivially Noetherian and the conjecture is false. We therefore use the standard notion from algebraic geometry over algebraic structures [1], which is stronger and makes the statement meaningful.

Definition 1. An *equation* over $M_n(K)$ in unknowns $X_1, \dots, X_k$ is a noncommutative polynomial in the X_l with coefficients in K and constants from $M_n(K)$; it is evaluated at a tuple of matrices in the obvious way. A system S of equations (of any cardinality) defines the *algebraic set* $V(S) \subseteq M_n(K)^k$ of its common zeros. $M_n(K)$ is *equationally Noetherian* if for every k and every system S there is a finite $S_0 \subseteq S$ with $V(S_0) = V(S)$; equivalently, algebraic sets satisfy the descending chain condition.

The conjecture asserts: for every field K , $M_n(K)$ is equationally Noetherian if and only if $n \leq 2$.

3 Result

Theorem 2. *For every field K and every n , $M_n(K)$ is equationally Noetherian. In particular $M_3(\mathbb{Q})$ is, and Conjecture 00000002058 is false.*

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4 Proof

Let $R = K[x_{l,i,j}]$ be the polynomial ring in kn^2 variables, one for each entry of each unknown. Substituting the *generic matrices* $(x_{l,i,j})_{i,j}$ for the unknowns and $(c_{ij}) \in M_n(R)$ for each constant $c \in M_n(K)$ gives an algebra homomorphism G from equations to $M_n(R)$. For every tuple of matrices X , the value of an equation p at X is obtained by evaluating the entries of $G(p)$ at the entries of X : both sides are algebra homomorphisms of p that agree on unknowns and constants. Hence

$$V(S) = \{X : G(p)_{ij}(X) = 0 \text{ for all } p \in S, i, j\}.$$

R is Noetherian (Hilbert’s basis theorem), so the ideal generated by the polynomials $G(p)_{ij}$ ($p \in S$) is generated by finitely many of them, coming from finitely many equations $p_1, \dots, p_r \in S$. If X is a common zero of $p_1, \dots, p_r$, then evaluation at X kills these generators, hence the whole ideal, hence every $G(p)_{ij}$. So $V(\{p_1, \dots, p_r\}) = V(S)$.

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `Equation K n k` — the free algebra over K on the unknowns `inl l` and the constants `inr A`, $A \in M_n(K)$; `value X` — evaluation, sending `inl l` to X_l and `inr A` to A ; `solutions S` — the algebraic set $V(S)$.
- `EqNoetherian K n` — the definition above; `ConjectureHolds` — for every field K (in `Type`) and every n , `EqNoetherian K n` $\leftrightarrow n \leq 2$.
- `generic`, `generic_eval` — the generic-matrix homomorphism and its compatibility with evaluation (by `FreeAlgebra.hom_ext`).
- `eqNoetherian` — Theorem 2, via Mathlib’s Noetherianity of multivariate polynomial rings and `Submodule.fg_span_iff_fg_span_finset_subset`.
- `conjecture_00000002058_false` — $\neg \text{ConjectureHolds}$, at $K = \mathbb{Q}$, $n = 3$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000002058_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The same argument shows that every finite-dimensional algebra over a field is equationally Noetherian. Infinite ascending chains of ideals of noncommutative polynomials (“commutator polynomials”) do exist in free algebras. But equational Noetherianity concerns the solution sets of equations, not ideals of the free algebra, and these stabilize because they are algebraic sets in an affine space over K .

References

- [1] G. Baumslag, A. Myasnikov and V. Remeslennikov, *Algebraic geometry over groups I. Algebraic sets and ideal theory*, J. Algebra 219 (1999), 16–79.